

Assignment 2 – 2nd-order modulator

- A. Design a 2nd-order **feed-forward** sigma delta modulator to digitize audio. The audio bandwidth is from 20Hz to 20kHz. [3 points]
- Determine the minimum OSR to achieve peak SQNR = 100dB, scan the input amplitude to find the peak SQNR.
 - Add thermal noise at the input of 1st integrator. Determine the noise level to achieve a peak SNR of 90dB. Scale the coefficients to keep both accumulators' output swing $< \pm 0.5$ V.
- B. Design a 2nd-order **feedback** sigma delta modulator with the same specifications as part A. [3 points]
- Determine the minimum OSR and the thermal noise to achieve a peak SNR of 90dB, SQNR = 100dB. Also scale the coefficients to keep both accumulators' output swing $< \pm 0.5$ V.
- C. Design a 2nd-order **feedback** sigma delta modulator with a notch in the signal bandwidth with the same specifications as above. (Optimize the notch to achieve minimum OSR with peak SNR = 90 dB, SQNR = 100 dB, also scale the coefficients.) [3 points]
- D. Discuss the difference between above three designs. [1 point]
- Assignment is due by **10:30am May 19th**
 - Instruction about how to submit assignment is on Brightspace
 - Please refer to the slides of practical2 for information about scaling and noise :D